

CBSE Class 12
Computer Science
Previous Year Question Paper 2014

Series: OSR

Code no. 91

- Please check that this question paper contains **16** printed pages.
- Code number given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **7** questions.
- **Please write down the Serial Number of the question before attempting it.**
- 15 minutes time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

COMPUTER SCIENCE

Time Allowed: 3 hours

Maximum Marks: 70

Instructions:

- 1) **All** questions are compulsory.
- 2) Programming Language: C++

1. (a) What is the difference between call by reference and call by value with respect to memory allocation? Give a suitable example to illustrate using C++ code. **2 Marks**

Ans:

Call by Value	Call by reference
----------------------	--------------------------

1. A variable's value is transferred to another variable.	1. A variable copy is made.
2. Example: <pre> Void main() { int a=2; b=3; cout<<a<<" "<<b<<endl; Implement(a,b); cout<<a<<" "<<b<<endl; } Void Implement(int a1, int b1) { a1++; b1 ++; cout<<a1<" "<<b1<<endl; } </pre> Output: 2 3 3 4 2 3	2. Example: <pre> Void main() { int a=2; b=3; cout<<a<<" "<<b<<endl; Implement(a,b); cout<<a<<" "<<b<<endl; } Void Implement(int a1, int b1) { a1++; b1 ++; cout<<a1<" "<<b1<<endl; } </pre> Output: 2 3 3 4 3 4

(b) Observe the following C++ code and write the name(s) of the header file(s), which will be essentially required to run it in a C++ compiler: 1 Mark

```

Void main ()
{
    char CH,STR[20];
    cin>>STR;
    CH=toupper (STR [0]);
    cout<<STR<<" starts with"<<CH<<endl;
}
  
```

Ans: <iostream.h> and ctype.h

(c) Rewrite the following C++ code after removing all the syntax error(s), if present in the code. Make sure that you underline each correction done by you in the code. 2 Marks

Important Note:

Assume that all the required header files are already included, which are essential to run this code.

The corrections made by you do not change the logic of the program.

```
typedef char[80] STR;
void main()
{
    Txt STR;
    Gets (Txt);
    cout<<Txt[0]<<'\t'<<Txt[2];
    cout<<Txt<<endl;
}
```

Ans: typedef char [80] STR;

```
Void main ()
{
    STR Txt;
    gets (Txt);
    cout<<Txt [0]<<'\t'<<Txt [2];
    cout<<Txt<<endl;
}
```

(d) Obtain the output from the following C++ program as expected to appear on the screen after its execution. 2 Marks

Important Note:

All the desired header files are already included in the which are required to run the code.

```
void main()
{
    char *Text="AJANTA";
    int *P,
    Num [] = {1, 5, 7, 9};
```

```
P=Num;
cout<<*P<<Text<<endl;
Text++;
P++;
cout<<*P<<Text<<endl;
}
```

Ans: Output:

1 AJANTA

5 JANTA

(e) Obtain the output of the following C++ program, which will appear on the screen after its execution. **3 Marks**

Important Note:

- All the desired header files are already included in the code, which are required to run the code.

```
class Game
{
    int Level, Score;
    char Type;
public:
    Game (char GType='P')
    {Level=1;Score=0;Type=GType;}
    void Play(int GS);
    void Change();
    void Show()
    {
        cout<<Type<<"@"<<Level<<endl;
        cout<<Score<<endl;
    }
};
```

```
void main ()
{
    Game A ('G'), B;
    B.Show ();
    A.Play (11);
    A.Change ();
    B.Play (25);
    A.Show ();
    B.Show ();
}

void Game :: Change()
{
    Type = (Type=='P')?'G':'P';
}

void Game :: Play (int GS)
{
    Score+=GS;
    If (Score>=30)
        Level=3;
    else if(Score>=20)
        Level=2;
    else
        Level=1;
}
```

Ans: Output:

P @ 1

0

P @ 1

11

P @ 2

25

(f) Read the following C++ code carefully and find out, which out of the given options (i) to (iv) are the expected correct output(s) of it. Also, write the maximum and minimum value that can be assigned to the variable Taker used in the code: 2 Marks

```
void main()
{
    int GuessMe [4]={100,50,200,20};
    int Taker=random(2)+2;
    for (int Chance=0;Chance<Taker; Chance++)
        cout<< GuessMe [Chance] <<"#";
}
```

(i) 100#

(ii) 50#100#

(iii) 100#50#200

(iv) 100#50

Ans: (iii) 100#50#200

Max value: 350

Min value: 170

2. (a) What is function overloading ? Write an example using C++ to illustrate the concept of function overloading. 2 Marks

Ans: Function overloading refers to when two or more functions have the same name but differ in the amount of parameters or the data types of the parameters.

Example:

```
Void sum (int a, int b)
```

```
{
```

```
    cout<<"Sum="<<a+b;
```

```

}
Void sum (int a, int b, int c)
{
cout<<"Sum="<<a+b+c;
}
Void main ()
{
int a=1, int b=2, c=3;
sum (a,b);
sum (a,b,c);
}

```

(b) Answer the questions (i) and (ii) after going through the following class
2 Marks

```

class Hospital
{
    int Pno, Dno;
public:
    Hospital (int PN);           //Function 1
    Hospital ();               //Function 2
    Hospital (Hospital &H);    //Function 3
    void In();                 //Function 4
    void Disp();              //Function 5
};
void main()
{
    Hospital H(20); //Statement 1
}

```

(i) Which of the functions out of Function 1, 2, 3, 4 or 5 will get executed when the Statement 1 is executed in the above code?

Ans: When Statement 1 is executed, Function 1 will be called.

(ii) Write a statement to declare a new object G with reference to already existing object H using Function 3.

Ans: Hospital G (H);

(c) Define a class Tourist in C++ with the following specification: 4 Marks

Data Members

- **CNo** - to store Cab No
- **CType** - to store a character 'A', 'B', or 'C' as City Type
- **PerKM** - to store per Kilo Meter charges
- **Distance** - to store Distance travelled (in KM)

Member Functions

- A constructor function to initialize CType as 'A' and CNo as '0000'
- A function CityCharges () to assign PerKM as per the following table:

CType	PerKM
A	20
B	18
C	15

- A function RegisterCab () to allow administrator to enter the values for CNo and CType. Also, this function should call CityCharges () to assign PerKM Charges.
- A function Display () to allow user to enter the value of Distance and display CNo, CType, PerKM, PerKM*Distance (as Amount) on screen.

Ans: class Tourist

```
{
    int CNo;
    Char CType;
    int PerKM;
    int Distance;
public:
```

```
Tourism ()
{
    CType='A';
    CNo=0;
}

void CityCharges ()
{
    if (CType=='A')
        PerKM=20;
    elseif (CType=='B')
        PerKM=18;
    elseif (Ctype=='C')
        PerKM=15;
}

void RegisterCab ()
{
    cout<<"Enter the CabNo'";
    cin>>CNo;
    cout<<"Enter Cab Type'";
    cin>>CType;
    CityCharges ();
}

Void display ()
{
    cout<<"Enter the distance";
    cin>>Distance;
    cout<<"Cab No is:"<<CNo<<endl;
    cout<<"Cab Type is:"<<CType<<endl;
```

```
cout<<" Per Kilometer charges is:"<<PerKM<<endl;
cout<<"Amount is:"<<PerKM*Distance;
}
};
```

(d) Consider the following C++ code and answer the question from (i) to (iv):
4 Marks

```
class University
{
    long Id;
    char City [20];
protected:
    char Country [20];
public:
    University ();
    void Register ();
    void Display ();
};
class Department : private University
{
    long DCode [10];
    char HOD [20];
protected:
    double Budger;
public:
    Department ();
    void Enter ();
    void Show ();
};
```

```
class Student : public Department
```

```
{
```

```
    long RollNo;
```

```
    char Name [20];
```

```
public:
```

```
    Student ();
```

```
    void Enroll ();
```

```
    void view ();
```

```
};
```

(i) Which type of Inheritance is shown in the above example?

Ans: In the example above, multilevel inheritance is demonstrated.

(ii) Write the names of those member functions, which are directly accessed from the objects of class Student.

Ans: Member function:

```
void Enroll ();
```

```
void View ();
```

```
void Enter ();
```

```
void Show ();
```

(iii) Write the names of those data members, which can be directly accessible from the member functions of class Student.

Ans: Data Members:

```
long RollNo;
```

```
char Name [20];
```

```
double Budget;
```

(iv) Is it possible to directly call function Display () of class University from an object of class Department? (Answer as Yes or No).

Ans: No, It's not possible since the Campus's DisplayO method is now private for Department class objects.

3. (a) Write a code for a function EvenOdd(int T[], int C) in C++, to add 1 in all the odd values and 2 in all the even values of the array T. 3 Marks

Example: If the original content of an array S is

T [0]	T [1]	T [2]	T [3]	T [4]
35	12	16	69	26

The modified content will be:

T [0]	T [1]	T [2]	T [3]	T [4]
36	14	18	70	28

Ans: void EvenOdd (int T [], int C)

```
{
    int i;
    for (i=0;i<=C-1;i++)
    {
        if (T [i]%2==0)
            T [i] = T[i] + 2;
        else
            T [i] = T[i] + 1;
    }
    cout<<"Modified content will be:-";
    for (i=0;i<=C-1;i++)
        cout<<T[i];
}
```

(b) An array A [20] [30] is stored along the row in the memory with each element requiring 4 bytes of storage. If the base address of array A is 32000, find out the location of A [15] [10]. Also, find the total number of elements present in this array. 3 Marks

Ans: B = 32000

W = 4

$$A [I] [J] = B + W [N (I - J_1) + (J + J_1)]$$

$$\begin{aligned} A [15] [10] &= 32000 + 4[30 (15-0) + (10-0)] \\ &= 32000 + 4[450 + 10] \\ &= 32000 + 4[460] \\ &= 32000 + 1840 \\ &= 33840 \end{aligned}$$

Location of A [10] [15] = 33840

Total number of elements present in this array = $20 \times 30 = 600$

(c) Write user-defined function Add End 2(int A [] [4], int N, int M) in C++ to find and display the sum of all the values, which are ending with 2 (i.e., units place is 2). For example if the content of array is: **2 Marks**

22	16	12
19	5	2

The output should be 36.

Ans: void AddEnd (int A [] [4], int N, int M)

```
{
    int i,j,Sum=0;
    for (i=0;i<=N-1;i++)
    {
        for (j=0;j<=M-1;j++)
        {
            if (A[i] [j] %10==2)
                Sum = Sum+ A [i] [j];
        }
    }
    cout<<Sum;
}
```

(d) Evaluate the following postfix expression. Show the status of stack after execution of each operation separately: **2 Marks**

T, F, NOT, AND, T, OR, F, AND

Ans:

SYMBOL	OPERATION	STACK
T	PUSH	T
F	PUSH	T,F
NOT	POP one element. Apply NOT	T,T
AND	POP two elements. Apply AND	T
T	PUSH	T,T
OR	POP two elements. Apply OR	T
F	PUSH	T,F
AND	POP two elements. Apply AND	F

(e) Write a function PUSHBOOK () in C++ to perform insert operation on a Dynamic Stack, which contains Book_no and Book_Title. Consider the following definition of NODE, while writing your C++ code. 4 Marks

struct NODE

```
{
int Book_No;
char Book_Title[20];
NODE *Next;
};
```

Ans: void POPBook ()

```
{
NODE*P = new NODE;
cout<<"Enter Book NO. and Book Title";
cin>>p->>Book_No;
gets (p->Book_Title);
if (top==Null)
    p ->Next= NULL;
    top= p;
```

```
else
{
    p ->Next= top;
    top =p;
}
}
```

4. (a) Fill in the blanks marked as Statement 1 and Statement 2, in the program segment given below with appropriate functions for the required task. **1 Mark**

```
class Agency
{
    int ANo; //Agent Code
    char AName[20]; //Agent Name
    char Mobile[12]; //Agent Mobile
public:
    void Enter(); //Function to enter details of agent
    void Disp(); //Function to display details of agent
    int RAno () {return ANo;}
    void UpdateMobile() //Function to update Mobile
    {
        cout<<"Updated Mobile:";
        gets (Mobile);
    }
};

Void AgentUpdate ()
{
    fstream F;
    F.open ("AGENT.DAT", ios:: binary|ios::in|ios::out);
```

```

int Updt=0;
int UAno;
cout<<"Ano (Agent No- to update Mobile):";
cin>>UAno;
Agency A;
while (!Updt && F.read((char*)&A,sizeof(A)))
{
    if (A.RAno()==UAno)
    {
        //Statement 1: To call the function to Update Mobile No.
        _____;
        //Statement 2: To reposition file pointer to re-write the updated object back
        in the file _____;
        F.write ((char*) & A,sizeof(A));
        Updt++;
    }
}
if (Updt)
cout<<"Mobile Updated for Agent"<<UAno<<endl;
else
cout<<"Agent not in the Agency"<<endl;
F.close ();
}

```

Ans: Statement 1: A. UpdateMobile ();

Statement 2: F.Seekg (-1*sizeof(A), ios :: cur);

(b) Write a function AECOUNT () in C++, which should read each character of a text file NOTES.TXT, should count and display the occurrence of alphabets A and E (including small cases a and e too). 2 Marks

Example:

If the file content is as follows:

CBSE enhanced its CCE guidelines further.

The AE Count () function should display the output as

A: 1

E: 7

Ans: void AECOUNT ()

```
{
Char Ch;
ifstream fcin ("NOTES.txt");
int count1= 0, count2=0;
while (!fcin.eof())
{
fcin.get(Ch);
if (Ch=='A' || Ch=='a')
count1 ++;
elseif (Ch=='E' || Ch=='e')
count2 ++;
}
fcin.close ();
cout<<"A:"<<count1<<endl;
cout<<"E:"<<count2<<endl;
}
```

(c) Assuming the class TOYS as declared below, write a function in C++ to read the objects of TOYS from binary file TOYS.DAT and display those details of those TOYS, which are meant for children of AgeRange "5 to 8".

3 Marks

class TOYS

```
{
```

int ToyCode;

```

char ToyName [10];
char AgeRange;
public:
void Enter ()
{
cin>>ToyCode;
gets (ToyName);
gets (AgeRange);
}
Void Display ()
{
cout<<ToyCode<<": "<<ToyName<<endl;
cout<<AgeRange<<endl;
}
char *WhatAge() {return AgeRange};
}
Ans: void Show ()
{
TOY T;
ifstream fcin ("TOY.DAT",ios :: in[ios :: binary);
while (fcin)
{
fcin.read ((char*) &T, sizeof (T));
if (stream(T.WhatAge(), "5 to 8"))
T. Display ();
}
fcin.close ();
}

```

5. (a) Explain the concept of Cartesian Product between two tables, with the help of appropriate example. 2 Marks

Ans:

TABLE: 1

Roll_No	Name	Marks
12	ABC	90
13	MNO	95
14	XYZ	85

TABLE: 2

SRoll_No	Name	Age
12	ABC	14
13	MNO	12
16	EPG	15

Select Roll_No, Name from Table1, Table2;

Roll_No	Name	Marks	SRoll_No	Name	Age
12	ABC	90	12	ABC	14
13	MNO	95	12	ABC	14
14	XYZ	85	12	ABC	14
12	ABC	90	13	MNO	13
13	MNO	95	13	MNO	13
14	XYZ	85	13	MNO	13
12	ABC	90	16	EPG	15
13	MNO	95	16	EPG	15
14	XYZ	85	16	EPG	15

NOTE: Answer the questions (b) and (c) on the basis of the following tables SHOPPE and ACCESSORIES.

TABLE: SHOPPE

ID	SName	Area
S01	ABC Computronics	CP
S02	All Infotech Media	GK II
S03	Tech Shoppe	CP
S04	Geeks Techno Soft	Nehru Place
S05	Hitech Store	Nehru Place

Table: ACCESSORIES

ID	IName	Price	SNo
A01	Mother Board	12000	S01
A02	Hard Disk	5000	S01
A03	Keyboard	500	S02
A04	Mouse	300	S01
A05	Mother Board	13000	S02
A06	Keyboard	400	S03
A07	LCD	6000	S04
A08	LCD	5500	S05
A09	Mouse	350	S05
A10	Hardisk	4500	S03

(b) Write the SQL queries:

4 Marks

(i) To display Name and Price of all the Accessories in ascending order of their Price.

Ans: select Name, Price from ACCESSORIES order by Price;

(ii) To display Id and S Name of all Shoppe located in Nehru Place.

Ans: select ID, S Name from SHOPPE where Area='Nehru Place';

(iii) To display Minimum and Maximum price of each Name of Accessories.

Ans: select MIN(Price), MAX(Price) from ACCESSORIES group by Name;

(iv) To display Name, Price of all Accessories and their respective SName where they are available.

Ans: select Name, Price, S Name' from SHOPPE, ACCESSORIES where SHOPPE.ID= ACCESSORIES.ID;

(c) Write the output of the following SQL commands: 2 Marks

(i) SELECT DISTINCT NAME FROM ACCESSORIES WHERE PRICE >= 5000;

Ans:

INAME
Mother Board
Hard Disk
LCD

(ii) SELECT AREA, COUNT (*) FROM SHOPPE GROUP BY AREA;

Ans:

AREA	Count(*)
CP	2
GK II	1
Nehru Place	2

(iii) SELECT COUNT (DISTINCT AREA) FROM SHOPPE;

Ans:

COUNT (DISTINCT AREA)
3

(iv) SELECT NAME, PRICE*0.05 DISCOUNT FROM ACCESSORIES WHERE SNO IN ('S02', 'S03');

Ans:

INAME	DISCOUNT
Keyboard	25
Motherboard	650
Keyboard	20

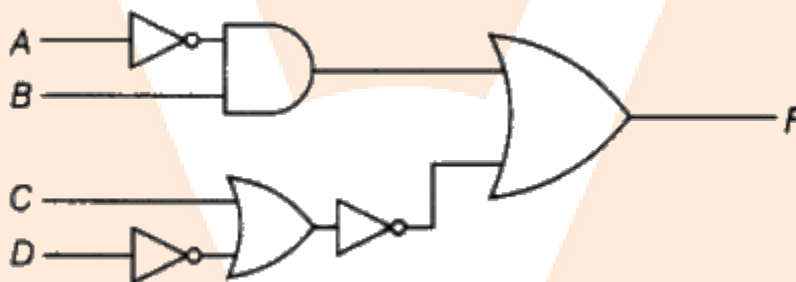
6. (a) Name the law shown below and verify it using a truth table. 2 Marks

$$X + X' \cdot Y = X + Y$$

Ans:

X	Y	X'O	X'Y	X+X'Y	X+Y
0	0	1	0	0	0
0	1	1	1	1	1
1	0	0	0	1	1
1	1	0	0	1	1

(b) Obtain the Boolean Expression for the logic shown below: 2 Marks



Ans: $F(A, B, C, D) = \underline{(A \cdot B)} + (\underline{C} + \underline{D})$

$$F(A, B, C, D) = \underline{(A \cdot B)} + (\underline{C} + \underline{D})$$

$$F(A, B, C, D) = \underline{(A \cdot B)} + (\underline{C} \cdot \underline{D})$$

(c) Write the Product of Sum form of the function $F(X, Y, Z)$ for the following truth table representation of F: 1 Marks

X	Y	Z	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0

1	1	0	1
1	1	1	1

Ans: $F(X, Y, Z) = (X+Y+\underline{Z}). (X+\underline{Y}+Z). (\underline{X}+Y+Z). (\underline{X} + Y + \underline{Z})$

(d) Obtain the minimal form for the following Boolean expression using Karnaugh's Map: **3 Marks**

$F(A,B,C,D) = \sum(1,3,4,5,6,7,12,13)$

Ans:

		C	D	C	D	C	D	C	D
		0	0	0	1	1	1	1	0
A	B								
0	0	0		1	1	3	1	2	
A	B								
0	1	4	1	5	1	7	1	6	1
A	B								
1	1	12		13	1	15		14	
A	B								
1	0	8		9	1	11		10	

Quad 1: $\underline{A}D$

Quad 2: $\underline{C}D$

Quad 3: $\underline{A}B$

Answer is $\underline{A}D + \underline{C}D + \underline{A}B$.

7. (a) Write two characteristics Wi-Fi.

1 Mark

Ans: Wi-Fi has the following characteristics:

- 1) It allows devices to connect to the network without using a wire.
- 2) A single internet connection can connect a group of devices.

(b) What is the difference between E-mail and Chat?

1 Mark

Ans: It is not necessary for the receiver to be online when sending an email, but it is required for the communicators to be online at the time of conversation in Chat.

(c) Expand the following:

1 Mark

- GSM
- GPRS

Ans: GSM: Global System for Mobile Communication

GPRS: General Packet Radio Service

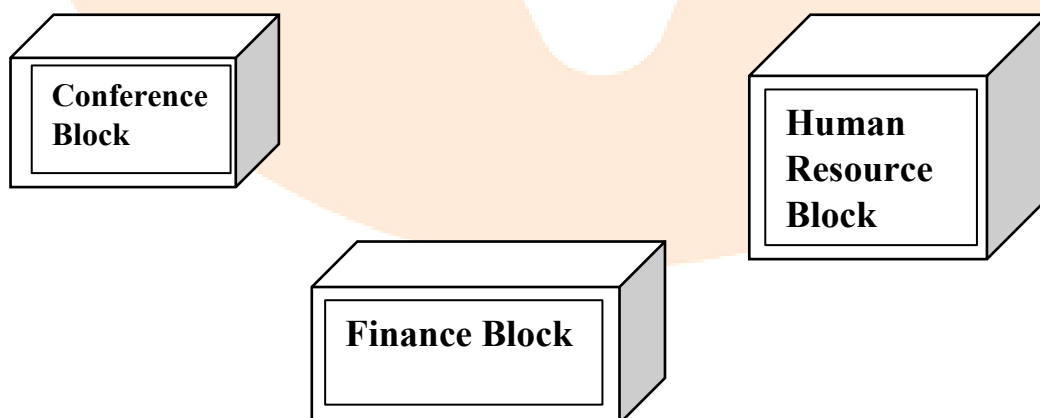
(d) Which type of network (out of LAN, PAN and MAN) is formed, when you connect two mobiles using Bluetooth to transfer a video.

1 Mark

Ans: PAN (Personal Area Network) is formed when you connect two mobiles using Bluetooth to transfer a video.

(e) Tech Up Corporation (TUC) is a professional consultancy company. The company is planning to set up their new offices in India with its hub at Hyderabad. As a network adviser, you have to understand their requirements and suggest to them the best available solutions. Their queries are mentioned as (i) to (iv) below.

Physical Locations of the blocks of TUC



Block to Block distances (in Mtrs.)

Block (From)	Block (T)	Distance
Human Resource	Conference	60
Human Resource	Finance	120
Conference	Finance	80

Expected Number of Computers to be installed in each block

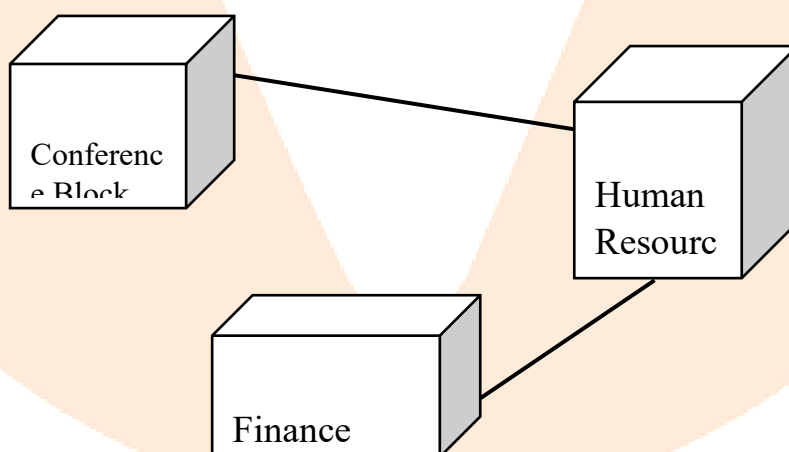
Block	Computers
Human Resource	125
Finance	25
Conference	60

(i) What will be the most appropriate block, where TUC should plan to install their server? **1 Mark**

Ans: The Human Resource Block is the best place to put the server.

(ii) Draw a block to block cable layout to connect all the buildings in the most appropriate manner for efficient communication. **1 Mark**

Ans:



(iii) What will be the best possible connectivity out of the following, you will suggest to connect the new setup of offices in Bangalore with its London based office? **1 Mark**

- Infrared
- Satellite Link

- **Ethernet Cable**

Ans: Satellite Link

(iv) Which of the following devices will be suggested by you to connect each computer in each of the buildings? 1 Mark

- **Gateway**
- **Switch**
- **Modem**

Ans: Switch

(f) Write names of any two popular Open Source Software, which are used as Operating Systems. 1 Mark

Ans: Linux and Unix are two widely used operating systems based on open source software.

(g) Write any two important characteristics of Cloud Computing. 1 Mark

Ans: Cloud Computing has two characteristics:

- (i) It is controlled by an entity and restricted to its approved users.
- (ii) It is available 24 hours a day, 7 days a week over the internet.